National Institute of Technology Kurukshetra

Department of Computer Engineering

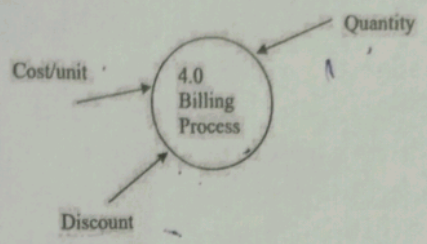
SOFTWARE ENGINEERING (ITPC 207/DSPC 207)

Date of Exam – 26-September-2025

Time: 08:00 AM-08:50 AM [50 min] **Max Marks:**

Note: All questions are compulsory.

1. The following portion of a DFD is not correct as (1)
   (a) there is no output data flow from the process
   (b) there are three data flow inputs to the process
   (c) there is no external entity
   (d) there is no data store ~~What is the need for test case prioritization? Explain different types of prioritization techniques.~~



2. Pick entities from the following: (1)
   (i) vendor
   (ii) student
   (iii) attends
   (iv) km/hour
   (a) i, ii, iii (b) i, ii, iv (c) i and ii (d) iii and iv
3. Which model is often used in large-scale software development where a modular approach is necessary? (1)
   A) Incremental model
   B) Spiral model
   C) V-Model
   D) RAD model
4. The SRS document is also known as _____ specification. (1)
   a) Black Box
   b) White Box
   c) Grey Box
   d) None
5. What is the major drawback of the Spiral Model? (1)
   a) Higher amount of risk analysis
   b) Doesn't work well for smaller projects
   c) Additional functionalities are added later on
   d) Strong approval and documentation control
6. Define SCM? What does "Win-Win" mean in the context of negotiating during the requirement engineering activity? (3)

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7. Discuss some of the problems that occur when requirements must be elicited from three or four different customers. How build and fix model different from the Incremental Model? Explain with the help of a neat diagram **(4)**

8. The Motor Vehicle Branch administers driving tests and issues driver's licenses. Any person who wants a driver's license must first take a learner's exam at any Motor Vehicle Branch in the province. If he/she fails the exam, he can take the exam again any time after a week of the failed exam date, at any branch. If he passes the exam, he is issued a license (type = learner's) with a unique license number. A learner's license may contain a single restriction on it. The person may take his driver's exam at any branch any time before the learner's license expiry date (which is usually set at six months after the license issue date). If he passes the exam, the branch issues him a driver's license. A driver's license must also record if the driver has completed driver's education, for insurance purposes. **(3)**

   Create a E-R diagram following these steps.

   1. Find out the entities and relationships.
   2. Figure out attributes of the entities and (if any) of the relationships.
   3. Figure out constraints between entities and relationships.